1)

Building	# of PCs	Subnet address (CIDR format)
1	3000	139.179.192.0/20
2	2200	139.179.208.0/20
3	2000	139.179.224.0/21
4	1500	139.179.232.0/21
5	1000	139.179.240.0/22

139.179.192.0/18

10001011 10110011 11000000 00000000

18 bits subnet part

↓ same for all 5 buildings

B1: $2^{11} < 3000 < 2^{12}$, 12 bits needed for host part ⇒ subnet = 32 − 12 = 20

B2: $2^{11} < 2200 < 2^{12}$, 12 bits needed for host part ⇒ subnet = 32 − 12 = 20

B3: $2^{10} < 2000 < 2^{11}$, 11 bits needed for host part ⇒ subnet = 32 − 11 = 21

B4: $2^{10} < 1500 < 2^{11}$, 11 bits needed for host part ⇒ subnet = 32 − 11 = 21

B5: $2^{9} < 1000 < 2^{10}$, 10 bits needed for host part ⇒ subnet = 32 − 10 = 22

B1: starts from initial ip address and occupies 4096 addresses ⇒ $\frac{4096}{256} = 16$

start: 139.179.192.0

next availability: 139.179.208.0
(192+16)

B2: start: 139.179.208.0

next availability: 139.179.224.0
(208+16)

(adding 16 again since $\frac{4096}{256}$)

B3: start: 139.179.224.0

next availability: 139.179.232.0
(224+8)

(adding 8 since $\frac{2048}{256}$)

B4: start: 139.179.232.0

next availability: 139.179.240.0
(232+8)

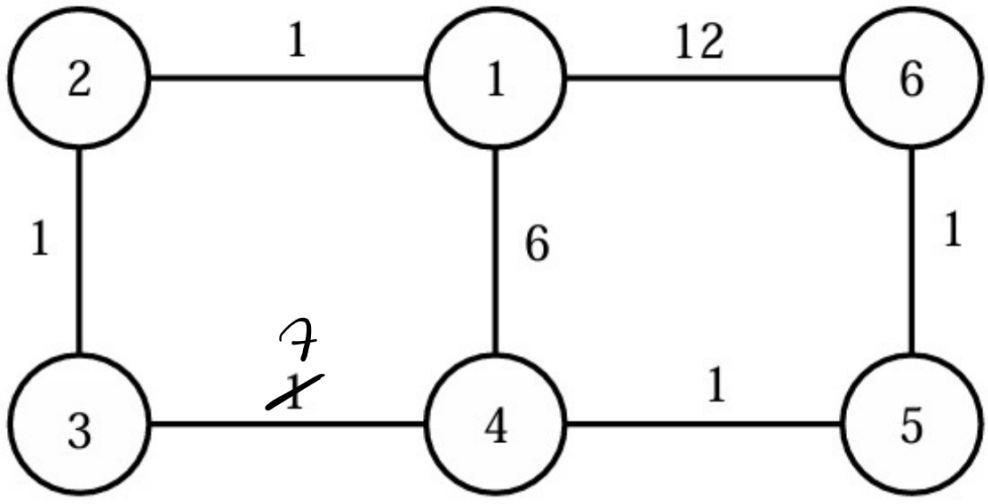
(adding 8 since $\frac{2048}{256}$)

B5: start: 139.179.240.0

next availability: 139.179.244.0
(240+4)

(adding 4 since $\frac{1024}{256}$)

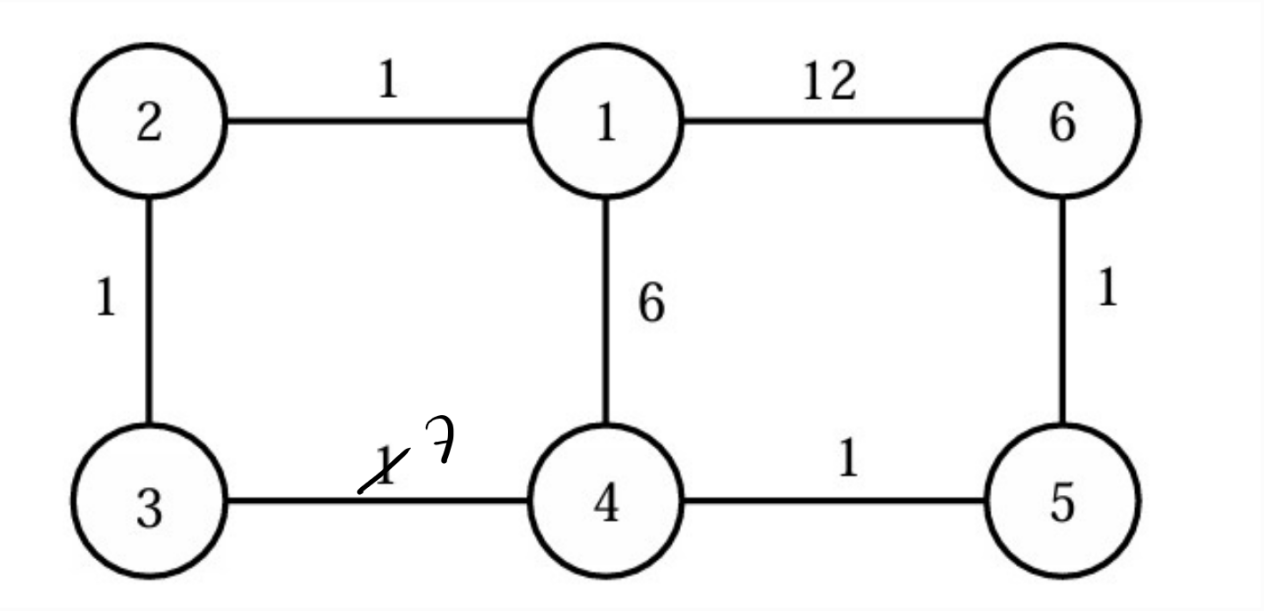
2)



Time, t	D^1 (6) via			D^2 (6) via		D^3 (6) via		D^4 (6) via			D^5 (6) via	
	2	4	6	1	3	2	4	1	3	5	4	6
0.1	5	8	12	6	4	5	3	11	4	2	3	1
0.5	5	8	12	6	4	5	9	11	10	2	3	1
1.1	5	8	12	6	6	5	9	11	12	2	3	1
2.1	7	8	12	6	6	7	9	11	12	2	3	1
3.1	7	8	12	8	8	7	9	13	14	2	3	1
4.1	9	8	12	8	8	9	9	13	14	2	3	1
5.1	9	8	12	9	10	9	9	14	16	2	3	1
6.1	10	8	12	9	10	10	9	14	16	2	3	1
7.1	10	8	12	9	10	10	9	14	16	2	3	1
8.1	10	8	12	9	10	10	9	14	16	2	3	1
9.1	10	8	12	9	10	10	9	14	16	2	3	1
10.1	10	8	12	9	10	10	9	14	16	2	3	1

stabilizes at 6.1

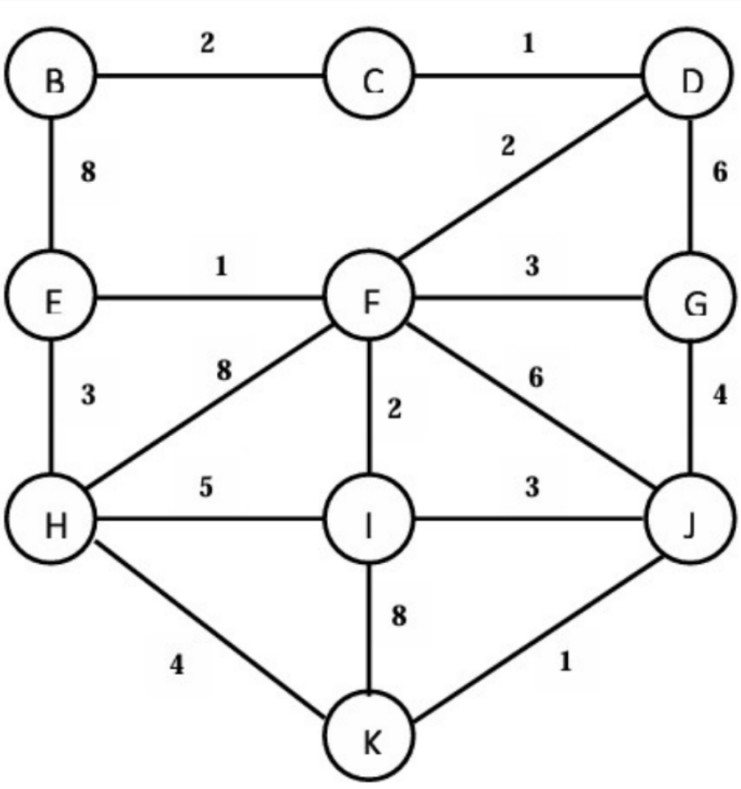
3) poisoned reverse



Time, t	D^1 (6) via			D^2 (6) via		D^3 (6) via		D^4 (6) via			D^5 (6) via	
	2	4	6	1	3	2	4	1	3	5	4	6
0.1	5	8	12	∞	4	∞	3	11	∞	2	∞	1
0.5	5	8	12	∞	4	∞	9	11	∞	2	∞	1
1.1	5	8	12	∞	10	∞	9	11	∞	2	∞	1
2.1	11	8	12	∞	10	∞	9	11	∞	2	∞	1
3.1	11	8	12	9	10	∞	9	∞	∞	2	∞	1
4.1	∞	8	12	9	10	10	9	∞	∞	2	∞	1
5.1	∞	8	12	9	10	10	9	∞	∞	2	∞	1
6.1												
7.1												
8.1												
9.1												
10.1												

stabilizes at $t = 6.1$

4)



Step	N'	~~D(C)~~, p(C)	~~D(D)~~, p(D)	~~D(E)~~, p(E)	~~D(F)~~, p(F)	~~D(G)~~, p(G)	~~D(H)~~, p(H)	~~D(I)~~, p(I)	~~D(J)~~, p(J)	D(K), p(K)
0	B	2,B	∞	8,B	∞	∞	∞	∞	∞	∞
1	BC	2,B	3,C	8,B	∞	∞	∞	∞	∞	∞
2	BCD	2,B	3,C	8,B	5,D	9,D	∞	∞	∞	∞
3	BCDF	2,B	3,C	6,F	5,D	8,F	13,F	7,F	11,F	∞
4	BCDFE	2,B	3,C	6,F	5,F	8,F	9,E	7,F	11,F	∞
5	BCDFEI	2,B	3,C	6,F	5,F	8,F	9,E	7,F	10,I	15,I
6	BCDFEJG	2,B	3,C	6,F	5,F	8,F	9,E	7,F	10,I	15,I
7	BCDFEIGH	2,B	3,C	6,F	5,F	8,F	9,E	7,F	10,I	13,H
	BCDFEIGHJ	2,B	3,C	6,F	5,F	8,F	9,E	7,F	10,I	11,J

5)

Destination Network	Next Hop
139.179.39.0 / 25	A
139.179.39.128 / 25	B
139.179.72.0 / 26	C
→ 196.101.153.64 / 26	D
139.179.0.0 / 16	E
→ 196.0.0.0 / 8	F

Destination IP Address	Next Hop	
139.179.39.130	B	
139.179.39.165	B	
139.179.72.66	E	can't be C since 66 > 63
196.101.153.127	D	
196.101.153.130	F	can't be D since 130 > 127

Rule : match with the longest matching prefix

A : 32 - 25 = 7 bits, $2^7 = 128$ ⇒ range : [0+0, 0+127] (last octet is zero)

B : 32 - 25 = 7 bits, $2^7 = 128$ ⇒ range : [128+0, 128+127]

C : 32 - 26 = 6 bits, $2^6 = 64$ ⇒ range [0+0, 0+63]

D : 32 - 26 = 6 bits, $2^6 = 64$ ⇒ range [64+0, 64+63]

E : 32 - 16 = 16 bits, $2^{16} = 65536$ ⇒ range [0+0, 0+65536]

F : 32 - 8 = 24, 2^{24} ⇒ range [0+0, 0+2^{24}]

6) D: 0110 1100110 , G: 110101

r = 5 ⇒ append 5 zeros to data sequence before appending.

appended: 0110 1100110 00000

```
110101
XOR
-----------------------
00001101100 00000
   110101
XOR
-----------------
0000 1100000
    110101
XOR
-----------
0001010
```

remainder ⇒ CRC bits: 01010

transmitted sequence: 01101100 1100 1010

error detection: we need $L < r$ for error detection, here we have $L = 4$ and $r = 5$, so $4 < 5$ holds. Meaning it can be detected

7) $R = 100\ Mbps = 10^8\ bps$, $s = 2\times10^8\ m/s$, $d_{max} = 400\ m$

we have: $F_{min} = \frac{2 d_{max} R}{s} \Rightarrow F_{min} = \frac{2 \times 400\ m \times 10^8\ bits/s}{2\times10^8\ m/s} = 400\ bits$

$\frac{400}{8} = 50\ bytes$

---

8) $R = 100\ Mbps = 10^8\ bps$, $m_A = 3$, $m_B = 4$, $m_C = 2$

$K_A \in \{0,1 \dots 7\}$, $K_B \in \{0,1 \dots 15\}$, $K_C \in \{0,1,2,3\}$

i) $Pr\{\text{suc retrans. by } c\} = Pr\{K_C < K_A \text{ and } K_C < K_B\}$

$$= \underbrace{\frac{1}{4}\cdot\frac{7}{8}\cdot\frac{15}{16}}_{\substack{c \text{ sends at } 0 \\ K_A>0 \\ K_B>0}} + \underbrace{\frac{1}{4}\cdot\frac{6}{8}\cdot\frac{14}{16}}_{\substack{c \text{ sends at } 1 \\ K_A>1 \\ K_B>1}} + \underbrace{\frac{1}{4}\cdot\frac{5}{8}\cdot\frac{13}{16}}_{\substack{c \text{ sends at } 2 \\ K_A>2 \\ K_B>2}} + \underbrace{\frac{1}{4}\cdot\frac{4}{8}\cdot\frac{12}{16}}_{\substack{c \text{ sends at } 3 \\ K_C>3 \\ K_C>3}}$$

$$\approx 0.59$$

ii) $Pr\{\text{collision of all nodes}\} = Pr\{K_A = K_B = K_C\}$

$$\frac{1}{4}\cdot\frac{1}{8}\cdot\frac{1}{16} \times 4 \approx 7.8\times10^{-3}$$

iii) $Pr\{K_A = K_B < K_C\}$

$K_C = 3 \Rightarrow K_A = K_B = 0,1,2 \rightarrow 3$ options

$K_C = 2 \Rightarrow K_A = K_B = 0,1 \rightarrow 2$ options

$K_C = 1 \Rightarrow K_A = K_B = 0 \rightarrow 1$ option

$$\frac{6}{512} \approx 0.012$$

$512 = 4\times8\times16$

9)

i) only Z will receive (find Z at port #3 from the switch table at A

ii) W, U, and T will receive (find U at port #4 from the switch table at B. Then, B will send everyone since it doesn't have U in its switch table) flooding

iii) X, Y, W, U, and T will receive (no T address found at the switch table at A → sends to X, Y, and B. B also floods W U T) floods

iv) B stored U and Y at step ii. So B forwards Y to switch. A stored Y at step iii. So A sends to Y (only Y receives)